

AI MOBILE APP FOR EARLY IDENTIFICATION OF STUDENT MENTAL HEALTH VIA BEHAVIOR & SPEECH

Project ID:25-26J-278

Project Proposal Report

B.Sc. (Hons) Degree in Information Technology.

Department of Information Technology

Sri Lanka Institute of Information Technology

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Declaration

I declare that this is my own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

Name	Student ID	Signature
KUMARASINGHE K D K Y	IT22321268	

The supervisor/s should certify the proposal report with the following declaration. The above candidates are carrying out research for the undergraduate Dissertation under my supervision.


Signature of the supervisor:

28/08/2025
Date

Signature of the Co-supervisor:

Date

Abstract

Psychological disorders such as depression, anxiety, and stress are rapidly increasing in university students, often undetected until they have reached severe stages. Traditional assessment methods that rely on self-reporting or clinical appointments are sporadic and reactive and fail to capture a critical opportunity for early and continuous surveillance. This project aims to develop a mobile app infused with AI that can predict early warning symptoms of upcoming mental health decline by analyzing multimodal emotional and behavioral information like tone of voice, screen behavior, typing behavior, sleep behavior, and mobility patterns.

The system collects passive data from the smartphone of a student and analyzes the data locally using privacy-preserving methods. Temporal deep learning architectures such as attention-augmented LSTMs and Temporal Convolutional Networks forecast individualized "windows of vulnerability" days or weeks in advance. Unlike conventional approaches using generalized thresholds, this system makes predictions for every student individually based on their behavioral baselines, lowering false alarms. Besides, explainable AI techniques such as SHAP values and attention-weight visualization provide human-readable explanations such as "Reduced consistency in sleep and enhanced negative emotionality this week may be an indicator of escalating emotional distress."

The intention is an anticipatory, privacy-preserving, and responsive solution that alerts students and clinicians prior to critical deterioration. Lastly, this initiative reframes mental health treatment from reactive crisis intervention to proactive prevention, enabling timely intervention and preventing long-term psychological harm in university students.

Keywords: Early Risk Prediction, Mental Health Decline, Temporal Deep Learning, Multimodal Behavioral Data, Explainable AI

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1. INTRODUCTION

Mental health conditions such as anxiety, depression, and stress-related disorders are increasingly prevalent among university students [1]. Academic pressures, social adjustments, and personal transitions often challenge their emotional well-being [2]. Despite rising concerns, identifying at-risk students and providing timely support remains difficult [3]. Traditional methods like self-report questionnaires and clinical interviews are subjective, infrequent, and reactive, highlighting the need for proactive, continuous monitoring of mental health indicators [4].

Speech analysis provides a reliable means of detecting emotional states [5]. Students experiencing depression or anxiety often show speech features such as flat tone, slower speaking rate, irregular rhythm, prolonged pauses, and limited pitch variation [6]. A mobile application can capture these features in real time, either during phone calls or diary-style audio recordings, offering insight into a student's emotional state [7]. Other behavioral signals, including screen time and typing patterns, further contribute to early detection [8]. Excessive or minimal device use may indicate withdrawal or compulsive avoidance, while slower typing, frequent edits, or negative language may reflect cognitive or emotional shifts [9].

Sleep patterns also play a key role. Monitoring device activity before sleep and after waking can detect irregular or fragmented sleep, which is associated with anxiety and depression [10]. By integrating speech, screen time, typing behavior, and sleep patterns, the system can generate a comprehensive profile of mental health [11]. Designed as an early identification tool rather than a diagnostic instrument, it provides timely alerts, self-care content, and guidance to seek professional support, addressing common barriers like stigma or limited access to care [12].

This study leverages artificial intelligence and mobile technology to develop a proactive mental health monitoring system [13]. Using time-series analysis and deep learning, it aims to detect subtle behavioral and emotional changes before noticeable symptoms emerge [14]. Ethical and privacy-aware AI ensures responsible monitoring, enabling students and healthcare professionals to intervene early [15]. By shifting from reactive to anticipatory care, the system

supports timely interventions and promotes overall well-being in higher education contexts [16].

1.1 Background and Literature survey

There are lots of studies that have been conducted about human depression, anxiety, and stress. The following studies specifically address Anomaly Detection for Stress and Mental Health.

One such study, titled *Early Detection of Anxiety, Depression, and Stress among Potential Patients using Machine Learning and Deep Learning Models* was conducted in 2023 by Alphonsa Sini P.J and Dr. Sherly K.K [17]. The research was outlined in the early detection of Anxiety, Depression, and Stress using ML and DL models. The information utilized in the study was obtained from Depression, Anxiety, and Stress Scales (DASS). Preprocessing of the data consisted of the imbalanced treatment using Synthetic Minority Oversampling Technique (SMOTE) and feature extraction using Random Forest algorithm. They specifically compare three significant algorithms — Support Vector Machine (SVM), Artificial Neural Network (ANN), and XGBoost — on a dataset with data relating to Anxiety, Depression, and Stress. The SVM model was most accurate in detecting mental health issues. Accuracy is 99.32% for Depression, accuracy is 99.80% for Anxiety, and accuracy is 98.44% for Stress.

In *Textual Sentiment Analysis for Mental Health Diagnosis* research done by Serra Aksoy in 2023 [18], the primary objective of this research was the use of sentiment analysis on text data, particularly from social media platforms like Twitter, Reddit, and TikTok to diagnose mental health conditions. The study provides a model that involves the integration of a Support Vector Classifier (SVC) and a Term Frequency-Inverse Document Frequency (TF-IDF) vectorizer for the application of sentiment analysis. The model achieves high accuracy for binary classification (depression/suicidal vs. normal) at 95.33% and for multiclass classification (depression, normal, suicidal) at 81.4%.

Accurate Human Stress Detection from Sleeping Habits Using a Hybrid Ensemble of Machine Learning Models by Thabitha S., Suresh A., and R. G. Rajeswari (ICDSAAI 2025) [19] proposes a solid framework that detects human stress from sleep-based features (sleep time, snore rate, respiration rate, body temperature, blood oxygen, limb/eye movement, heart rate, etc.). The authors build a hybrid ensemble of Logistic Regression, Gradient Boosting, Random Forest, AdaBoost, and a Support Vector Classifier using soft-voting in order to leverage the strengths of multiple learners. With a Kaggle sleep dataset, correct preprocessing, and an 80/20 train-test split, the ensemble significantly outperforms individual models — with an overall

accuracy of 99.21% and AUC of 0.9945 — and the project comes with a Streamlit demo for real-time user input and stress prediction.

1.2 Research Gap

The current methods for assessing mental health risks in students predominantly rely on self-reported questionnaires, clinical interviews, or one-time screenings [20]. These approaches are highly subjective and may not capture the actual mental state of students [21]. For example, a student may underreport symptoms due to stigma or fear of being judged, while another may exaggerate responses to gain attention or support. This reliance on self-assessment limits reliability and can lead to misdiagnosis or delayed interventions [21].

Another gap is that existing AI-based studies focus mainly on detection at a single point in time rather than predicting future deterioration [22]. While speech and text-based sentiment analysis models have shown promising results in identifying depression, anxiety, or stress [22], they often classify the current state only. This reactive approach fails to anticipate early warning signs of decline and does not provide enough time for proactive care before the condition worsens [23].

Furthermore, many models operate in controlled environments or with curated datasets (e.g., social media posts, lab-recorded speech, or survey-based DASS data), which may not accurately reflect the real-world behaviors of students in daily life [23]. Continuous, passive monitoring through mobile devices — such as screen usage, typing dynamics, and sleep patterns — remains underexplored in predictive modeling, leaving a significant gap in translating AI research into practical, real-time tools for student populations [24].

Another limitation is the lack of integration of multi-modal data streams [24]. Most existing works examine speech, text, or sleep data in isolation [24]. However, mental health deterioration is often reflected through interconnected behavioral patterns (e.g., irregular sleep paired with reduced social interactions and negative speech sentiment) [24]. The absence of holistic, time-series modeling prevents systems from generating a reliable risk profile over time [24].

Finally, ethical and privacy considerations remain under-addressed [24]. Many predictive systems lack transparency in risk scoring, explainability of outputs, or safeguards for sensitive personal data [24]. Without privacy-aware and explainable AI frameworks, adoption in real-world educational settings becomes difficult [24].

Source/ Approach	Data Type Used	Temporal Modeling	Real- world Monitorin g	Multi- modal Integratio n	Ethical / Privacy	Gap Relevant
Self-reported questionnaires / interviews	Subjective survey data	No (snapshot only)	No	No	Limited anonymity only	Highly subjective; stigma, exaggeration, or underreporting affect reliability
Existing AI-based studies (speech/text sentiment analysis)	Speech or text data	Mostly classification at one time point	Limited (lab or social media datasets)	No	Not transparent in outputs	Focuses on current state, not future risk prediction
Lab-controlled AI experiments	Curated datasets (lab speech, surveys)	Sometimes time-bound	No (not daily-life monitoring)	Rare	Weak privacy considerations	Lack of real-world applicability
Mobile usage studies	Behavioral logs	Limited or isolated	Sometimes explored	Rarely integrated	Limited safeguards	Underexplored predictive modeling in

(screen time, sleep, typing)						real-life context
Existing predictive models (few works)	Single data stream (e.g., speech or sleep)	Limited forecasting	No continuous monitoring	Not holistic	Rare Explainability	Cannot generate reliable risk profile
General predictive systems in education	Varies	Varies	Weak	Weak	Weak	Lack privacy-aware, explainable frameworks

Table 1 1.2Research Gap Comparison

1.3 Research Problem

Individuals experiencing early signs of mental health decline often go unnoticed until symptoms become severe, as traditional methods rely on self-reporting or infrequent clinical checkups [25]. This reactive approach prevents timely support and increases the risk of serious consequences [26]. While some mobile apps exist to track sleep, mood, or activity, they focus on single signals and cannot fully capture the complex, multi-layered changes that occur before mental health deterioration [27]. Moreover, these systems rarely adapt to personal behavior patterns and often generate results that are difficult to understand or trust [28].

Another key issue is the lack of integration across multiple behavioral signals—such as speech, sleep, mobility, screen usage, and sentiment—into one predictive and user-friendly system [28]. Current tools do not provide clear, explainable insights that individuals and clinicians can act on, leaving early warning opportunities unused [29]. As a result, there is a major gap in proactive mental health care: people lack a reliable, personalized, and interpretable system that can forecast their risk of decline days or even weeks in advance [30]. The proposed research aims to address this gap by developing a multi-modal, transparent, and adaptive deep learning framework that detects early warning signs from everyday behavioral data, enabling preventative care rather than delayed interventions [30].

2.OBJECTIVES

2.1 Main Objectives

The general objective of this project is to design an AI module capable of forecasting early warning signs of future decline in mental well-being by making use of multimodal behavioral and affect data such as tone of voice, sleep, screen use, and keyboard behavior. Employing temporal deep learning models over these continuously collected signals, the system shall be designed to identify personalized "windows of vulnerability" weeks or days before symptom appearance that are detectable clinically. By offering timely, interpretable warnings and risk predictions, the solution facilitates proactive self-care behavior among students and enables healthcare professionals to respond before crises arise, thereby shifting mental health care from reactive diagnosis towards preventive care.

2.2 Specific Objectives

Three specific objectives must be reached to achieve the overall objective described below.

1. Collection and processing of multimodal behavioral and emotional data.

The system will gather speech tone, typing cadences, sleep patterns, screen time, and mobility cues from the user's device. The raw streams of data will be pre-processed into structured summaries capable of tracking day-to-day emotional and behavioral changes without divulging sensitive personal information.

2. Prediction of personalized risk windows using temporal deep learning models.

The processed data will be fed into temporal deep learning models such as attention-augmented LSTMs and Temporal Convolutional Networks. These will make individualized "windows of vulnerability" predictions for each user with forecasts for possible mental health deterioration days or weeks before noticeable symptoms.

3. Generation of timely and interpretable risk alerts.

Based on the estimated risk windows, the system will provide early warnings to users in the form of clear and comprehensible alerts (e.g., "Decreased sleep consistency and

increased negative sentiment this week may indicate elevated risk"). The alerts will enable both users and clinicians to take preventive action to prevent further decline.

3. Methodology

- To facilitate early detection of mental health deterioration, this module makes use of an sufficient amount of multi-modal streams of data accumulated non-invasively and passively from user devices. These kinds of data sources provide continuous, real-time awareness regarding users' emotional, behavioral, and physiological states:

- **Speech Data (Tone, Emotion, and Sentiment)**

Speech serves as a vital indicator of emotional well-being. The system analyzes paralinguistic features such as:

- **Pitch** (voice frequency), which may indicate anxiety or excitement when elevated.
- **Tension**, reflecting stress levels through vocal strain.
- **Volume** (loudness) and **Speech Rate**, where abrupt changes can signal emotional shifts.
- **Pauses** or hesitations, which may correlate with cognitive load or emotional distress.

Sentiment analysis is applied to spoken content to detect the underlying emotional tone:

- **Positive Sentiment** (e.g., "I'm feeling hopeful") reflects well-being.
- **Negative Sentiment** (e.g., "I'm overwhelmed") may indicate early signs of decline.
- **Neutral Sentiment** offers context and behavioral baselines.

- **Textual Data (Messages and Sentiment Trends)**

The text language processing from messages, notes, or diaries provides semantic information about mood change, rumination, or avoidance. Natural Language Processing (NLP) models detect emotional tone, keywording behaviors (e.g., "exhausted," "can't get through"), and temporal polarity shifts in sentiment.

- **Sleep Behavior**

Irregular sleep will typically happen prior to mental health disorder. Sensor fusion techniques, such as the one by Sohrab Saeb (2017), use this variable to predict primary sleep periods from smartphone data streams—ambient light, activity records, audio, screen state, battery state, and communication records. With up to 92% accuracy with bespoke models, it identifies sleep onset, sleep length, and sleep regularity for single users.

Screen Usage and Typing Behavior

Excessive or uncontrolled screen time, nighttime phone use, and changes in typing patterns (i.e., rate, error rates, pressure distributions) can be indicative of anxiety, insomnia, or mood instability. All of these web-related behaviors are monitored continuously for the detection of potential indicators of cognitive overload or emotional distress.

3.1 System Architecture

The proposed system architecture enables proactive, privacy-aware, and real-time prediction of mental health risks for university students. Figure 3.1 illustrates the high-level flow:

1. **Data Collection Layer** – The mobile application passively detects multimodal behavioral and affective data, including tone of speech, sleep, screen time, typing habits, and indicators of mobility. The students can also record mood logs or self-reports through the application interface.
2. **Preprocessing & Summarization Layer** – On-device AI models preprocess raw data and create concise summaries, ensuring user privacy by preventing raw personal data from leaving the device.
3. **Prediction Layer** – Temporal deep learning models such as Attention-augmented LSTMs and Temporal CNNs analyze behavioral summaries to forecast “windows of vulnerability” days or weeks in advance.
4. **Alert & Explanation Layer** – Customized alerts are generated when risk scores exceed individual thresholds. Explainable AI (e.g., SHAP values) translates model decisions into human-understandable explanations, e.g., "Your sleep duration declined 30%, and negative sentiment increased this week."
5. **Intervention Layer** – Reinforcement learning agents deliver context-aware, real-time interventions such as mindfulness prompts or self-care recommendations.

Table 2 3.1 Technologies, techniques, architectures, and algorithms used.

Category	Choices Used
Technologies	Python, TensorFlow Lite, PyTorch Mobile, React Native, Flask/FastAPI
Techniques	Temporal Deep Learning, Federated Learning, Explainable AI (XAI)
Algorithms	Attention-Augmented LSTM, Temporal CNN, SHAP-based Interpretability
Architectures	On-device Inference Architecture, Federated Learning Architecture

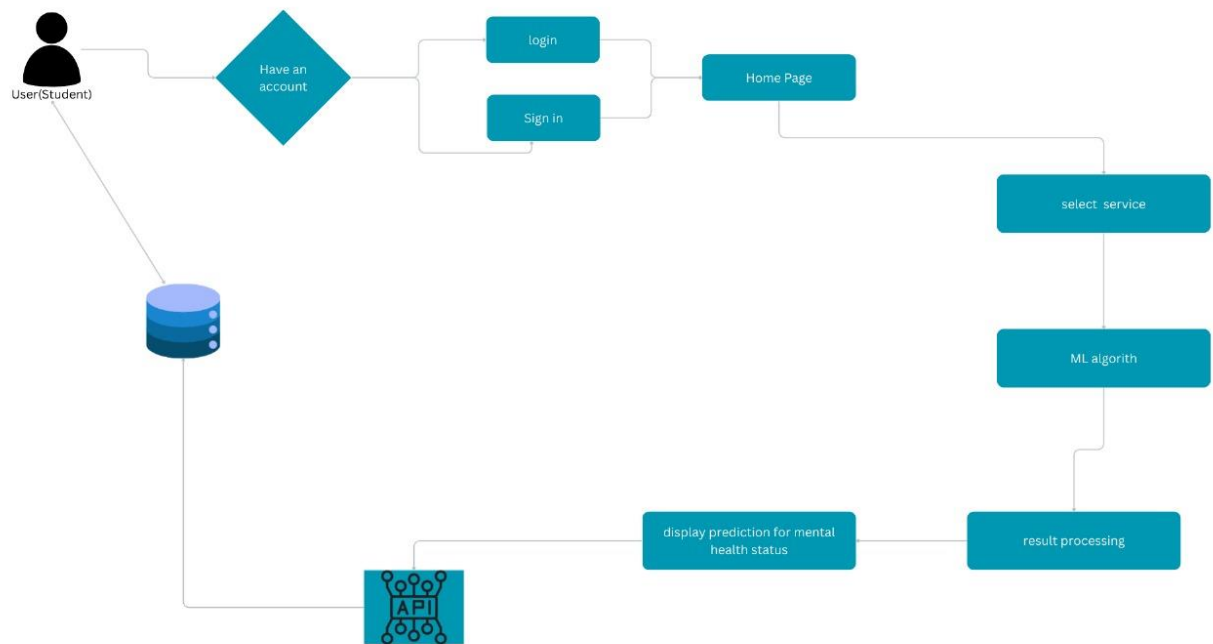


Figure 1 3.1Individual system diagram

3.2 Software Solution

The proposed solution will follow the **Agile methodology** with **Scrum** to ensure iterative development and continuous feedback.

Development Phases:

- **Requirement Gathering:** Collection of behavioral and emotional datasets, including speech tone, screen usage, and sleep patterns; surveys with students for baseline data.
- **Feasibility Study:**
 - Economic Feasibility – Low-cost, mobile-based solution with no need for expensive hardware.
 - Technical Feasibility – Expertise in temporal deep learning, mobile app development, and on-device AI deployment.
 - Scheduling Feasibility – Deliverable within the academic project timeline with pilot testing.
- **System & Software Design:**
 - Mobile App UI – React Native interface for data input, alerts, and recommendations.
 - Model Backend – Temporal deep learning models running on-device using TensorFlow Lite or PyTorch Mobile.
 - Database – Encrypted SQLite for secure on-device storage.
 - APIs – Federated learning framework for decentralized model updates while preserving privacy.

Project Requirements:

- **Functional:** Risk prediction, real-time alerts, explainable outputs, personalized interventions.
- **Non-Functional:** Privacy, scalability, reliability, high availability, user-friendliness.

- System Requirements: Android/iOS compatibility, lightweight on-device inference, encrypted local storage.

3.3 Commercialization

This research is suggested to develop a mobile application for the early identification of deteriorating student mental health with AI-driven behavior and speech analysis. The target market is university students, counselors, universities, and mental health agencies. There will be a basic and a premium version of the system. The basic version will provide behavioral monitoring and alerts based on tone of speech, typing behavior, screen usage, and sleep habits. The premium version will have higher capabilities like personalized risk forecasting, intervention supported by reinforcement learning, and explainable analytics. The application is user-friendly, does not require any technical knowledge in advance, and ensures privacy by performing data processing on-device. Our workforce comprises experienced members with backgrounds in AI, mobile development, and mental health frameworks. The funding requirement is estimated to be 200–300 USD per month, and the revenue will come in the form of freemium subscriptions and institutional licensing to universities and counseling centers. Marketing will be done through university collaborations, online advertising, and awareness campaigns for the purpose of inducing adoption by students and institutions.

4. Project requirements

4.1 Functional Requirements

1. The system should collect multimodal behavioral and emotional data from students' smartphones, including speech tone, screen usage, typing behavior, sleep patterns, and mobility signals.
2. The system should preprocess and summarize the collected data using on-device AI to ensure privacy before analysis.
3. The system should apply temporal deep learning models (e.g., attention-augmented LSTMs, Temporal CNNs) to predict early risk windows for future mental health decline.
4. The system should continuously update predictions as new behavioral data becomes available, refining accuracy over time.
5. The system should generate proactive alerts to students when the risk of mental health deterioration crosses a personalized threshold.
6. The system should provide explainable AI outputs so that users and clinicians can understand the reasons behind elevated risk scores.
7. The system should maintain privacy by processing all data on-device and ensuring no raw personal data is transmitted to external servers.

4.2 Non-functional Requirements

1. **User-Friendliness** – The application should have a clean, intuitive interface and be available on both Android and iOS platforms with a consistent experience across devices.
2. **Privacy & Security** – The system must comply with data protection standards (e.g., GDPR, HIPAA) by using on-device processing, encryption, and federated learning methods.
3. **Performance** – The system should provide real-time risk analysis with minimal latency and accurate predictions based on multimodal data inputs.
4. **Reliability & Availability** – The application should be available 24/7 with minimal downtime, ensuring continuous monitoring and timely alerts.
5. **Scalability** – The architecture should support many users simultaneously without performance degradation

4.3 System Requirements

Required Software:

1. VS code - To implement the code in python.
2. Python – For developing and training temporal deep learning models and backend services.
3. Flutter - To design a cross platform mobile based application..

4.4 User Requirements

This mobile application will be developed for multiple user groups:

Students – Students can view their mental health risk levels through simple, proactive alerts and access self-care recommendations before a critical decline occurs.

5. Grantt Chart

Month	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec-Jan	Feb	Mar	Apr	May
Brainstorming session														
Topic Selection														
Supervisor and Co-supervisor selection														
Group Registration														
Create TAF Doc														
TAF Submission														
Create Log Books														
Create Proposal Presentation														
Proposal Presentation														
Create Proposal Report														
Proposal Report Submission														
UI Designing														
Implementation of work														
Training Data Sets														
Working on ML Algorithms														
PP1 Creation (Progress Presentation 1)														
PP1														
Writing Research Paper														
Research paper submission														
PP2 Creation (Progress Presentation 2)														
PP2														
Create Final Report														
Final Report submission														
Testing the application														
Create Project Status Document														
Create Final Presentation														
Final Presentation and VIVA														

Figure 25.1 Grantt Chart

5.1 Work Breakdown Structure (WBS)

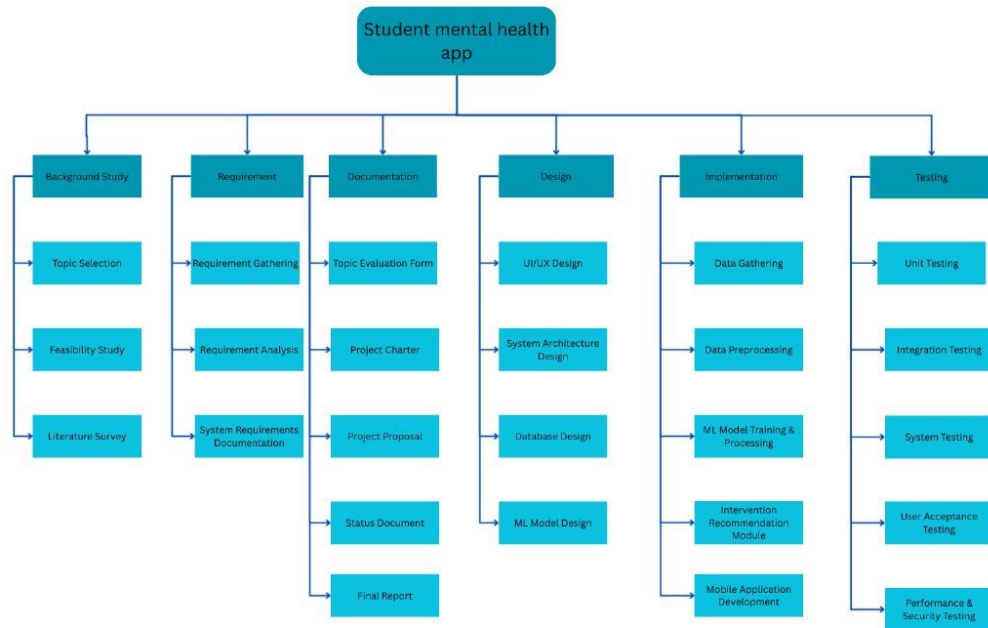


Figure 3 5.2Work Breakdown Structure

6. BUDGET AND BUDGET JUSTIFICATION

Expenses	Requirement Cost (Rs.)	Justification
Travelling cost for surveys and awareness programs	15,000.00	To conduct initial data collection, distribute questionnaires, and raise awareness about burnout among students at universities.
Data Collection (standardized questionnaires, survey tools, digital forms)	10,000.00	Required for acquiring reliable datasets from students for training, testing, and validating the system.
Deployment cost (server hosting, backend services)	10,796.00 / month	To maintain system availability and reliability through cloud-based deployment.
Hosting cost in Play Store	8,076.00	One-time cost to publish the mobile application on the Google Play Store for student access.
Miscellaneous (printing, promotional activities, student awareness sessions)	5,000.00	For marketing, leaflets, and awareness campaigns to ensure successful adoption of the app.
Total Expenses	48,872.00 + 10,796.00/month	—

Table 3 Table 6.1 provides an overall budget for the overall proposed system

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